

HEIDELBERG, Germany

WMO: 107345

Lat: 49.393N

Lon: 8.652E

Elev: 109

StdP: 100.02

Time Zone: 1.00 (EUC)

Period: 90-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-5.8	-12.2	1.3	-6.0	-10.0	1.6	-3.7	10.8	8.6	9.3	8.4	1.9	60	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	32.1	20.4	30.2	19.7	28.2	18.7	21.6	29.0	20.7	27.9	19.8	26.5	3.1	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.1	23.9	18.2	13.2	23.3	17.2	12.5	22.3	63.6	29.3	60.3	27.8	57.4	26.6	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.1	6.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	2.7	4.0	7.6	11.1	15.8	18.2	21.5	21.0	16.0	11.2	6.6	3.4
	DBStd	7.55	4.57	4.62	3.48	3.82	3.58	3.41	3.60	3.46	2.87	3.97	3.59	3.93
	HDD10.0	871	228	172	88	31	2	0	0	0	36	108	207	
	HDD18.3	2721	485	403	332	218	94	44	11	15	81	223	351	464
	CDD10.0	1467	1	2	14	64	181	247	357	341	179	72	7	2
	CDD18.3	276	0	0	0	2	14	42	109	97	10	1	0	0
	CDH23.3	2497	0	0	0	29	137	341	1047	857	82	4	0	0
	CDH26.7	845	0	0	0	3	19	84	412	310	18	0	0	0
(14) Wind	WSAvg	2.6	2.9	2.7	3.0	2.9	2.7	2.6	2.5	2.4	2.3	2.3	2.3	2.8
(15) Precipitation	PrecAvg	796	59	57	57	51	77	73	81	67	63	68	71	73
	PrecMax	1040	123	133	157	132	176	122	168	207	129	191	126	165
	PrecMin	549	4	17	13	3	18	19	25	7	15	14	4	11
	PrecStd	118	30	30	34	28	38	28	35	42	30	36	31	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.7	16.2	20.1	26.1	29.0	30.9	34.9	35.0	29.8	23.8	15.9	13.8
		MCWB	10.3	10.7	12.6	15.1	18.0	19.7	20.8	21.0	18.4	17.5	11.9	11.0
	2%	DB	12.0	13.9	17.1	23.8	26.2	29.0	32.9	32.1	25.2	20.8	14.0	12.1
		MCWB	9.5	9.5	10.8	14.2	16.6	19.1	21.2	20.4	17.3	16.0	11.1	10.2
	5%	DB	10.2	12.0	15.1	21.2	24.9	27.1	31.0	30.0	23.2	19.0	12.9	11.1
		MCWB	8.0	8.4	10.0	13.0	15.8	18.2	19.9	19.8	16.4	14.8	10.4	8.8
	10%	DB	9.1	10.1	13.2	18.8	23.0	25.0	29.0	28.0	21.2	17.2	11.8	9.8
		MCWB	7.4	7.6	9.2	11.9	14.7	17.4	19.3	18.6	15.6	13.9	9.6	8.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.6	13.6	16.3	19.7	22.0	23.0	22.9	19.5	17.8	12.9	12.0
		MCDB	13.1	15.5	18.6	24.1	27.1	26.9	31.7	30.8	25.2	22.3	14.2	13.0
	2%	WB	9.9	10.1	11.8	14.5	17.7	20.1	21.7	21.6	18.3	16.7	11.6	10.8
		MCDB	11.6	12.8	15.7	22.1	24.1	26.7	29.8	28.8	23.0	19.8	13.5	12.2
	5%	WB	8.6	9.0	10.8	13.2	16.6	19.2	20.8	20.7	17.5	15.6	10.7	9.3
		MCDB	10.3	11.6	14.0	19.8	22.7	25.3	28.6	27.6	21.8	18.3	12.4	10.9
	10%	WB	7.3	7.8	9.8	12.3	15.7	18.2	20.0	19.7	16.5	14.4	9.8	7.8
		MCDB	8.8	9.8	13.1	18.1	21.2	23.9	27.1	26.0	20.5	16.9	11.2	9.4
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	6.3	7.7	9.5	10.2	9.3	10.2	10.5	9.2	7.7	5.2	4.3
		MCDBR	6.1	9.8	11.5	13.4	13.9	12.9	13.6	14.1	12.9	10.4	6.6	5.5
	5% WB	MCWBR	4.9	6.3	6.6	6.1	6.3	5.5	4.8	5.4	6.2	6.2	4.7	4.5
		MCDDBR	6.0	8.4	10.0	12.4	12.1	11.5	12.2	12.2	10.3	9.0	6.2	5.4
(40) Clear-Sky Solar Irradiance	taub	taub	0.327	0.355	0.388	0.418	0.426	0.429	0.431	0.426	0.397	0.379	0.350	0.324
		taud	2.387	2.325	2.246	2.203	2.215	2.241	2.230	2.269	2.323	2.358	2.381	2.384
	Ebn at Noon	Ebn at Noon	716	778	811	826	835	833	825	810	795	740	676	662
		Edh at Noon	67	89	114	132	135	133	133	122	105	86	67	59
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.85	1.62	2.79	4.30	5.10	5.70	5.38	4.66	3.42	2.01	0.98	0.67
	RadStd	RadStd	0.13	0.26	0.40	0.56	0.55	0.46	0.53	0.43	0.41	0.21	0.10	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page